Results outline

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##   year pgsGSL leafout coloredLeaves gdddiff pdsi tmeanmax tmeanmean tmeanmin
## 1 2016    156     124           280    1594 -4.0       29        23       17
## 2 2017    156     117           273    1472  1.6       27        22       16
## 3 2018    144     123           267    1569  1.9       28        23       17
## 4 2019    143     119           262    1549  4.0       28        22       17
## 5 2020    133     125           258    1645 -2.4       29        23       17
## 6 2021    144     116           260    1543  1.0       28        23       17
## 7 2022    137     121           258    1611 -2.7       29        23       17
## 8 2023    145     113           258    1459 -0.3       27        22       16
## 9 2024    134     118           252    1573  2.3       28        23       17
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1 Wildchrokie

1.1 Growing season length and conditions

1. 2020 was the shortest season by days (mean GSL =135.9), with an average summer temperature of 23.1°C,and a summer thermal sum of 1644.6 GDD.
   2019 was the longest season by days (mean GSL = 141.5) with an average summer temperature of 22.4°C, and a summer thermal sum of 1549.4 GDD.
   2018 had an average GSL of 138.7 days, an average summer temperature of 22.9°, and summer thermal sum of 1569.5 GDD

2. Leafout dates across all years and species varied from 112 to 157 and budset date varied from 225 to 297, which are the two phases that set the boundaries of the growing season.

3. 2019 had the earliest start of season (mean leafout date = 121.7), and the coldest spring of the three years (8.7°C), with a spring thermal sum of 491.1 GDD.
   2020 had the latest start of season (mean leafout date = 141), and was also the warmest spring (9.1°C), but had the lowest thermal sum of three years (458.1 GDD).
   2018 had an average start of season of 127.3,an average spring temperature of 8.9°, and the highest spring thermal sum (511 GDD).

4. On average, the earliest leafout date was in 2019 for all species (*A. incana* : 120.9, B alleghaniensis: 123, *B. papyrifera* : 125.8 and *B. populifolia* : 121.5). The latest leafout date for all species was in 2020 (*A. incana* : 149.1, B alleghaniensis: 141.3, *B. papyrifera* : 136.6 and *B. populifolia* : 133.9). No species was consistently the earliest nor the latest to leafout.

5. 2019 had the earliest end of season (mean budset date =263.2), and was coldest autumn (average autumn temperature = 8.7°C), with a autumn thermal sum of 491.1 GDD.
   2020 had the latest end of season (mean budset date =276.8), and was the warmest autumn (average autumn temperature = 9.1°C) with a autumn thermal sum of 458.1 GDD.
   2018 had an average end of season of 266, an average autumn temperature of 8.9°, and had the highest thermal sum of the three years (511 GDD).

6. On average, the earliest budset date was in 2019 for *A. incana* (263.3) and *B. alleghaniensis* (258), but in 2018 for *B. papyrifera* (266.8) and *B. populifolia* (266.3). The latest budset date for all species was in 2020 (*A. incana* : 280.9, *B. alleghaniensis* : 264.1, *B. papyrifera* : 279.4 and *B. populifolia* : 282). No species was consistently the earliest nor the latest to set buds.

7. The driest spring of the three years (MAM) (3-month averaged PDSI index) was 2020 with a PDSI value of 1.3, a cumulative precipitation of 275.3 mm. 2019 was the least dry spring with a PDSI value of 2.7 and a cumulative precipitation of328.4 mm.

8. 2020 had the driest summer (JJA) with a PDSI value of -2.4, and a cumulative precipitation of 199.1 mm. 2019 was the least dry summer with a PDSI value of 4 and a cumulative precipitation of 346 mm.

9. 2020 is the only year a with period of moderate drought ($PDSI[-3, -2]$) which was from June through August. Throughout the three years, September of 2020 was the only period with a severe drought ($PDSI < -3$)

1.2 Ring widths

1. The ring widths of our species vary from 0.4 mm to 13.5 mm.

2. Ring widths for *B. alleghaniensis* were the smallest and varied from 0.4 mm to 11.1 mm, and *B. populifolia* had the largest rings which spanned from 0.5 mm to 2 mm on average. *A. incana* (0.9 mm to 4 mm) and *B. papyrifera* were quite similar (0.5 mm to 5.7 mm).

1.3 Model results

1. Growth did not change with greater thermal season for most species *A. incana* : (β = 0.1, UI_{90}[0, 0.2]), *B. alleghaniensis* : (β = 0.1, UI_{90}[0, 0.2], and *B. populifolia* : (β = 0, UI_{90}[-0.2, 0.2]]. However, *B. papyrifera* grew more with longer thermal season (β = 0.3, UI_{90}[0, 0.6])) (Figure 1 a) and e)).

2. In contrast, most species grew more with longer calendar seasons (*A. incana* : (β = 0.1, UI_{90}[0.1, 0.2]), *B. alleghaniensis* : (β = 0.1, UI_{90}[0, 0.2]) and *B. papyrifera* : (β = 0.3, UI_{90}[0.1, 0.5]). Similarly to its response to the thermal season, *B. populifolia* did not grow more with longer calendar seasons: (β = 0, UI_{90}[-0.1, 0.1]) (Figure 1 b) and f).

3. Growth decreased with a later start of season (SOS) for two species *A. incana* (β = -0.1, UI_{90}[-0.2, -0.1]) and *B. populifolia* (β = -0.1, UI_{90}[-0.2, 0]), but didn't change for the other two (*B. alleghaniensis* (β = 0, UI_{90}[-0.1, 0]) and *B. papyrifera* (β = -0.1, UI_{90}[-0.2, 0])) (Figure 1 c) and g)). The estimates were unchanged when using the full or restricted datasets (n =239 vs n =227).

4. Growth decreased with a later end of season (EOS) only for for *A. incana* (β = -0.1, UI_{90}[-0.2, 0]). Growth for the other species did not change with a later end of season (*B. populifolia* showed a weaker response but appears to also grow less with a later EOS (β = -0.1, UI_{90}[-0.2, 0], *B. alleghaniensis* (β = 0.1, UI_{90}[0, 0.2]) and *B. papyrifera* (β = 0.1, UI_{90}[0, 0.3]) (Figure 1 d) and g)). The estimates were unchanged when using the full or restricted datasets (n =230 vs n =227 Figure **??**).

5. Growth did not change across any sites (S. Hypolyte (α_{site} = 0.1, UI_{90}[-0.8, 0.9]), D. College (α_{site} = -0.1, UI_{90}[-0.9, 0.7]), W. Mountains (α_{site} = 0.1, UI_{90}[-0.7, 1]) and H. Forest (α_{site} = = -0.2, UI_{90}[-1, 0.7]) (Figure 2))

   Our results show similar trends in response to the thermal (GDD) (Figure S5) versus the calendar growing season (GSL) (Figure S6), with a stronger response to GSL. Specifically, *A. incana* shows a marked shift in its response strength with longer GSL compared to GDD.

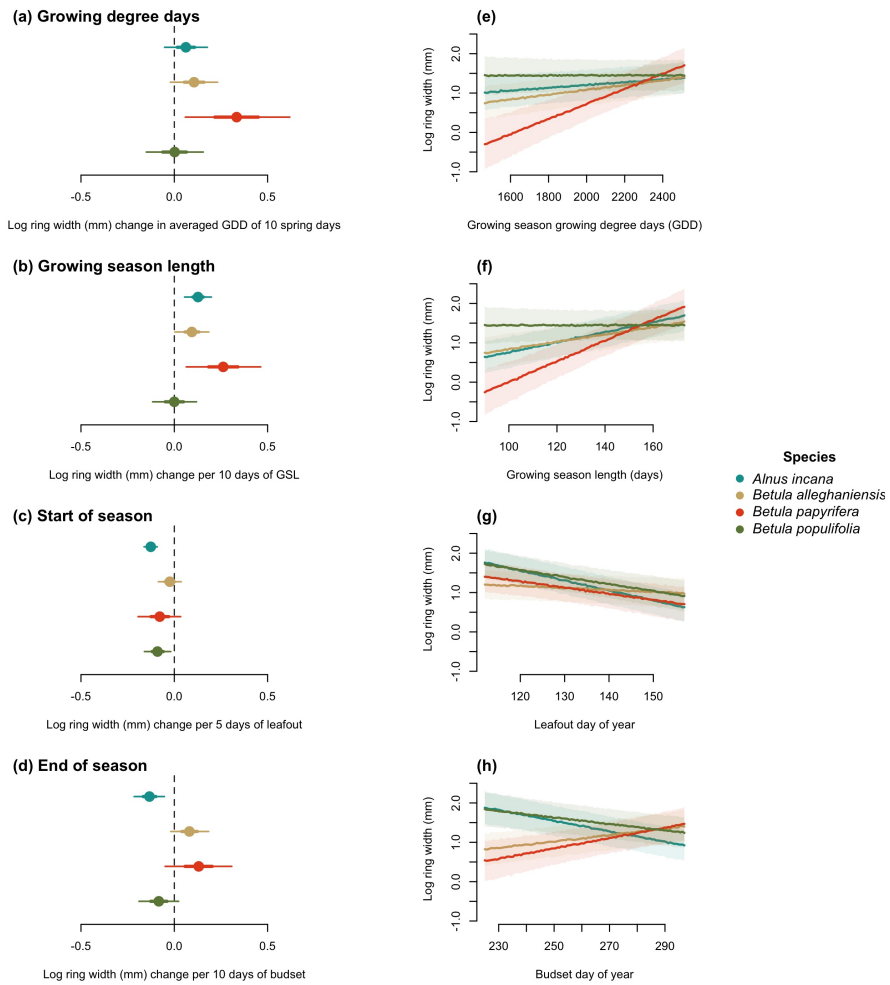

Figure 1: Model slope parameters across the different model predictors. GDD represent the accumulated growing degree days between leafout and budset, GSL the number of days between these phases. SOS represent the start of season, which is leafout and EOS, the end of season through budset. All predictors were fit using the same model and same observation, with only the independent variable changing (see Figure **??** for a comparison between two different data set length for SOS, and Figure S1, for end of EOS). The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

2 CoringTreespotters

2.1 Species, years and observations

1. CoringTreespotters includes eleven deciduous tree species with phenological observations from 2016 to 2024. In total, our final dataset has 239 leafout observations and 230. We used these observations to set the boundaries of individual-level growing seasons, which together yield 280 complete growing season lengths (GSL).

2.2 Growing season length and conditions

1. Across our nine years, 2023 had the lowest summer thermal sum (1458.9 GDD), and 2020 had the highest summer thermal sum (1644.6 GDD).

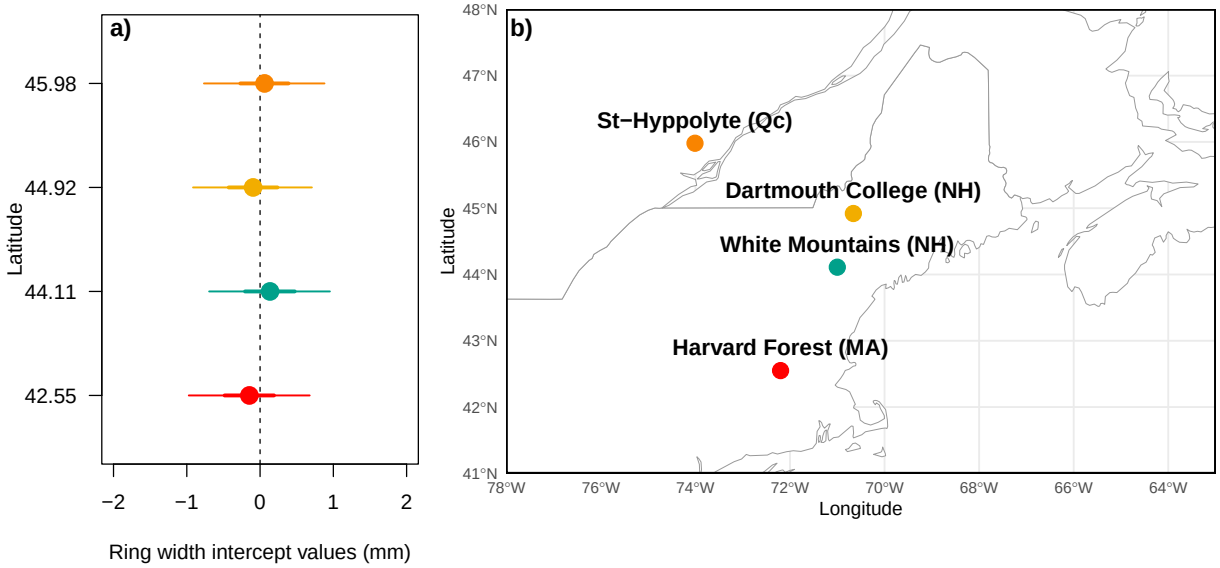

Figure 2: The effect of sites on growth across a latitudinal gradient. a) Model intercept parameters for each site with growing degree days as the predictor. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

2. 2020 was the shortest season by days (mean GSL =133.2), with an average summer temperature of 23.1°C, and a summer thermal sum of 1644.6 GDD.
   2017 was the longest season by days (mean GSL = 156) with an average summer temperature of 21.5°C, and a summer thermal sum of 1471.7 GDD.

3. Leafout dates across all years and species varied from 98 (2021) to 136 (2020) and leaf colouring date varied from 159 (2018) to 311 (2016).

4. 2023 had the earliest start of season (mean leafout date = 113.5), an average temperature of (10.1°C), and a spring thermal sum of 530.8 GDD.
   2020 had the latest start of season (mean leafout date = 125.2), an average temperature of (9.1°C), and a thermal sum of (458.1 GDD). This year coincides with the shortest growing season and had the fourth earliest EOS 258.4.

5. On average, the earliest leafout date for each species are *A. rubrum*: 104 (2023), *A. saccharum*: 111 (2017), *A. flava*: 102.7 (2016), *B. alleghaniensis* : 104 (2017), *B. nigra*: 108 (2017), *C. glabra*: 110 (2017), *C. ovata*: 117.8 (2020), *P. deltoides*: 122.5 (2018), *Q. alba*: 112 (2017), *Q. rubra*: 118.2 (2020), *T. americana*: 109 (2017),

6. 2024 had the earliest end of season (mean leaf colouring date = 251.8), an average temperature of (13.1°C), and an autumn thermal sum of 531.1 GDD.

   2016 had the latest end of season (mean leaf colouring date = 279.6), an average temperature of (13.2°C), and an autumn thermal sum of (521.4 GDD).

7. The driest spring of the three years (MAM) (3-month averaged PDSI index) was 2020 with a PDSI value of 1.3, a cumulative precipitation of 275.3 mm. 2019 was the least dry spring with a PDSI value of 2.7 and a cumulative precipitation of328.4 mm.

8. 2020 had the driest summer (JJA) with a PDSI value of -2.4, and a cumulative precipitation of 199.1 mm. 2019 was the least dry summer with a PDSI value of 4 and a cumulative precipitation of 346 mm.

9. 2020 is the only year a with period of moderate drought ($PDSI[-3, -2]$) which was from June through August. Throughout the three years, September of 2020 was the only period with a severe drought ($PDSI < -3$)

2.3 Ring widths

1. The ring widths of our species vary from 0.4 mm to 11.7 mm.

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2.4 Model results

1. Growth did not change with greater thermal season for most species (*A. rubrum*: $\beta = 0$, UI_{90}[-0.2, 0.1], *A. saccharum*: $\beta = 0.1$, UI_{90}[-0.1, 0.2], *A. flava*: $\beta = 0$, UI_{90}[-0.1, 0], *B. nigra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. glabra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. ovata*: β = -0.1, UI_{90}[-0.2, 0], *P. deltoides*: $\beta = 0$, UI_{90}[-0.1, 0], *Q. alba*: $\beta = 0.1$, UI_{90}[-0.2, 0.3], *Q. rubra*: $\beta = 0$, UI_{90}[-0.2, 0.2], and *T. americana*grew more with longer thermal season $\beta = 0$, UI_{90}[-0.1, 0]). However, growth for *B. alleghaniensis* decreased with greater thermal season β = -0.3, UI_{90}[-0.4, -0.1])) (Figure 3 a) and e)).

2. Similarly to the response to thermal season, the species did not increase growth with longer calendar season (*A. rubrum*: $\beta = 0.1$, UI_{90}[-0.1, 0.2], *A. saccharum*: $\beta = 0$, UI_{90}[-0.1, 0.1], *A. flava*: $\beta = 0$, UI_{90}[0, 0], *B. nigra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. glabra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. ovata*: $\beta = 0$, UI_{90}[-0.1, 0], *P. deltoides*: $\beta = 0$, UI_{90}[-0.1, 0], *Q. alba*: $\beta = 0.1$, UI_{90}[-0.1, 0.2], *Q. rubra*: $\beta = 0$, UI_{90}[-0.2, 0.1], and *T. americana*: $\beta = 0$, UI_{90}[-0.1, 0])) , including *B. alleghaniensis* : β = -0.1, UI_{90}[-0.2, 0] (Figure 3 b) and f)).

3. Growth did not change in response to the start of season (*A. rubrum*: β = -0.1, UI_{90}[-0.3, 0], *A. saccharum*: $\beta = 0$, UI_{90}[0, 0.1], *A. flava*: $\beta = 0$, UI_{90}[0, 0.1], *B. alleghaniensis* : β = -0.1, UI_{90}[-0.2, 0], *B. nigra*: $\beta = 0$, UI_{90}[-0.1, 0], *C. glabra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. ovata*: $\beta = 0$, UI_{90}[-0.1, 0.1], *P. deltoides*: $\beta = 0$, UI_{90}[-0.1, 0.1], *Q. alba*: β = -0.1, UI_{90}[-0.2, 0.1], *Q. rubra*: $\beta = 0$, UI_{90}[-0.1, 0.1], and *T. americana*: $\beta = 0.1$, UI_{90}[0, 0.1]) (Figure 3 c) and g)).

4. Similarly to the response to thermal season, the species did not increase growth with longer calendar season (*A. rubrum*: $\beta = 0$, UI_{90}[-0.1, 0.2], *A. saccharum*: $\beta = 0$, UI_{90}[-0.1, 0.1], *A. flava*: $\beta = 0$, UI_{90}[0, 0], *B. nigra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. glabra*: $\beta = 0$, UI_{90}[-0.1, 0.1], *C. ovata*: $\beta = 0$, UI_{90}[-0.1, 0], *P. deltoides*: $\beta = 0$, UI_{90}[-0.1, 0], *Q. alba*: $\beta = 0.1$, UI_{90}[-0.1, 0.2], *Q. rubra*: $\beta = 0$, UI_{90}[-0.1, 0.1], and *T. americana*: $\beta = 0$, UI_{90}[-0.1, 0])) , except *B. alleghaniensis* for which its growth decreased with later end of season (β = -0.1, UI_{90}[-0.2, 0]) (Figure 3 b) and f))

Supplemental results

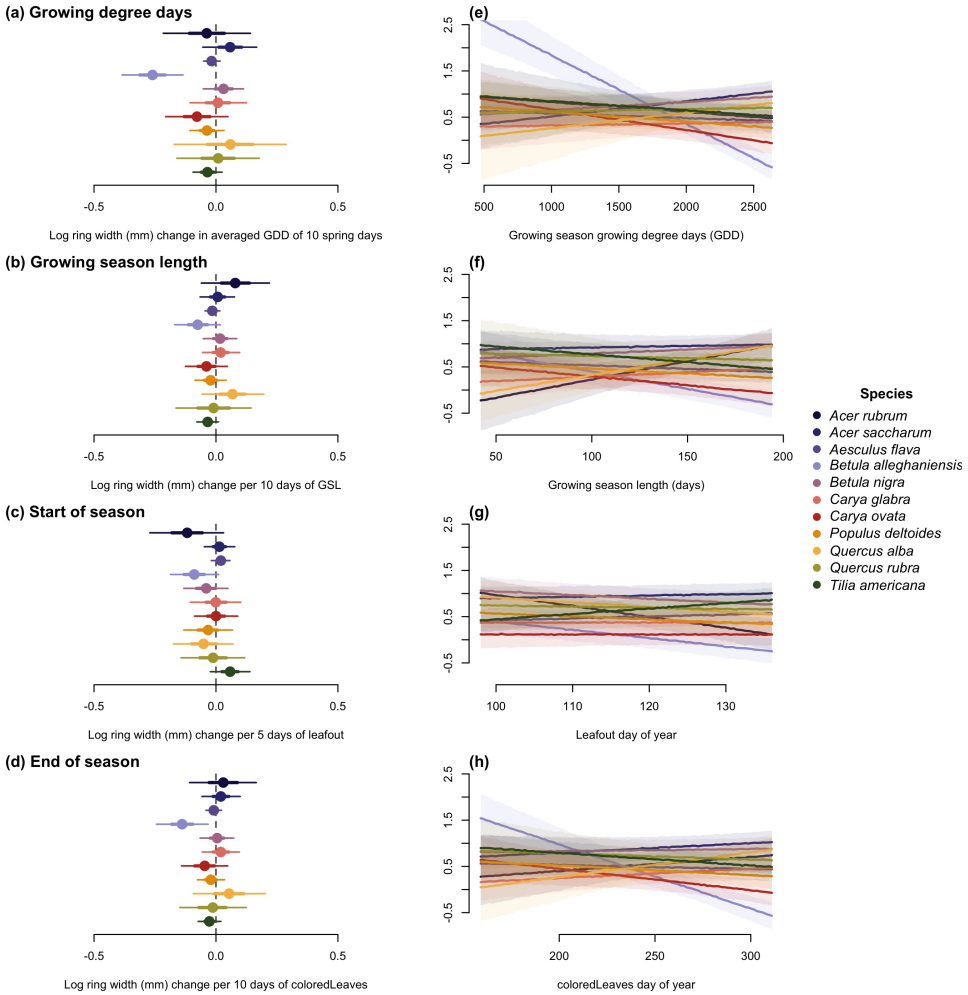

Figure 3: Model slope parameters across the different model predictors. GDD represent the accumulated growing degree days between leafout and budset, GSL the number of days between these phases. SOS represent the start of season, which is leafout and EOS, the end of season through budset. All predictors were fit using the same model and same observation, with only the independent variable changing (see Figure **??** for a comparison between two different data set length for SOS, and Figure S1, for end of EOS). The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.

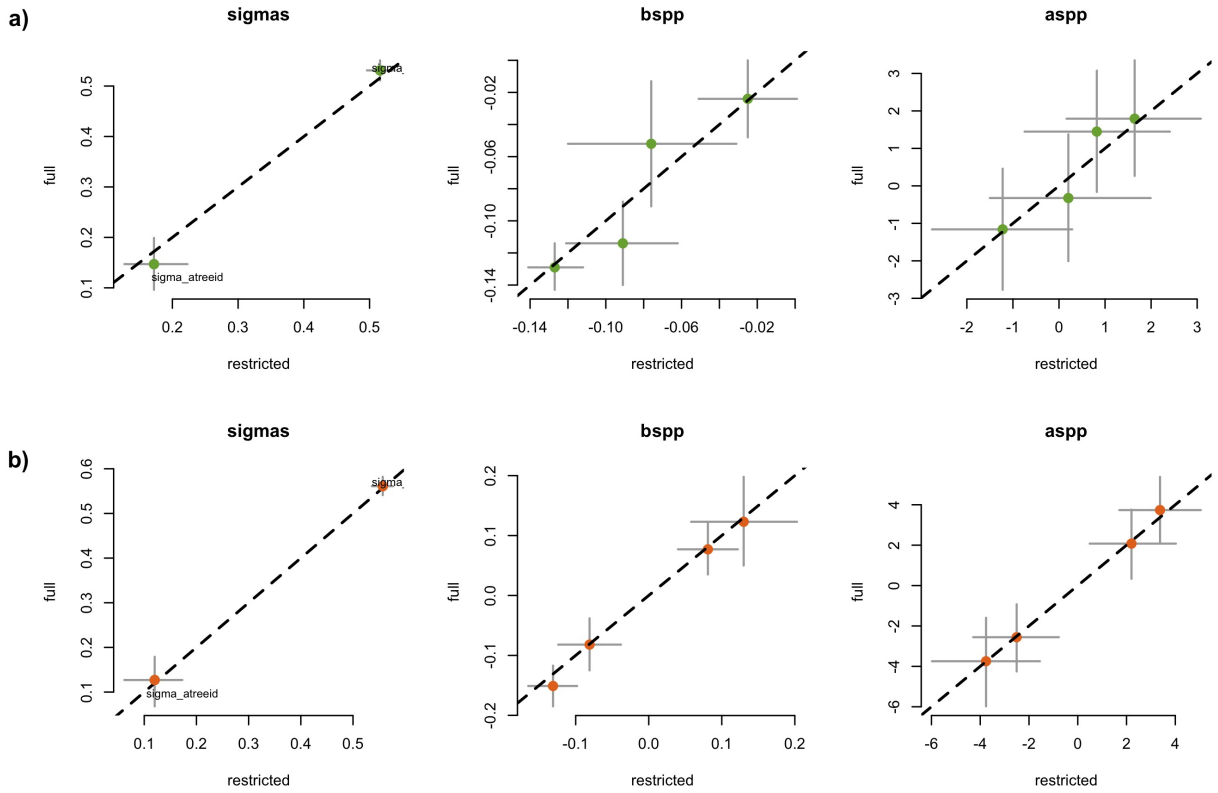

Figure S1: Full vs most restricted rows of data for the model with SOS (a)) EOS (b)) as the predictor. Each pannel represent the different parameters used to fit the models.

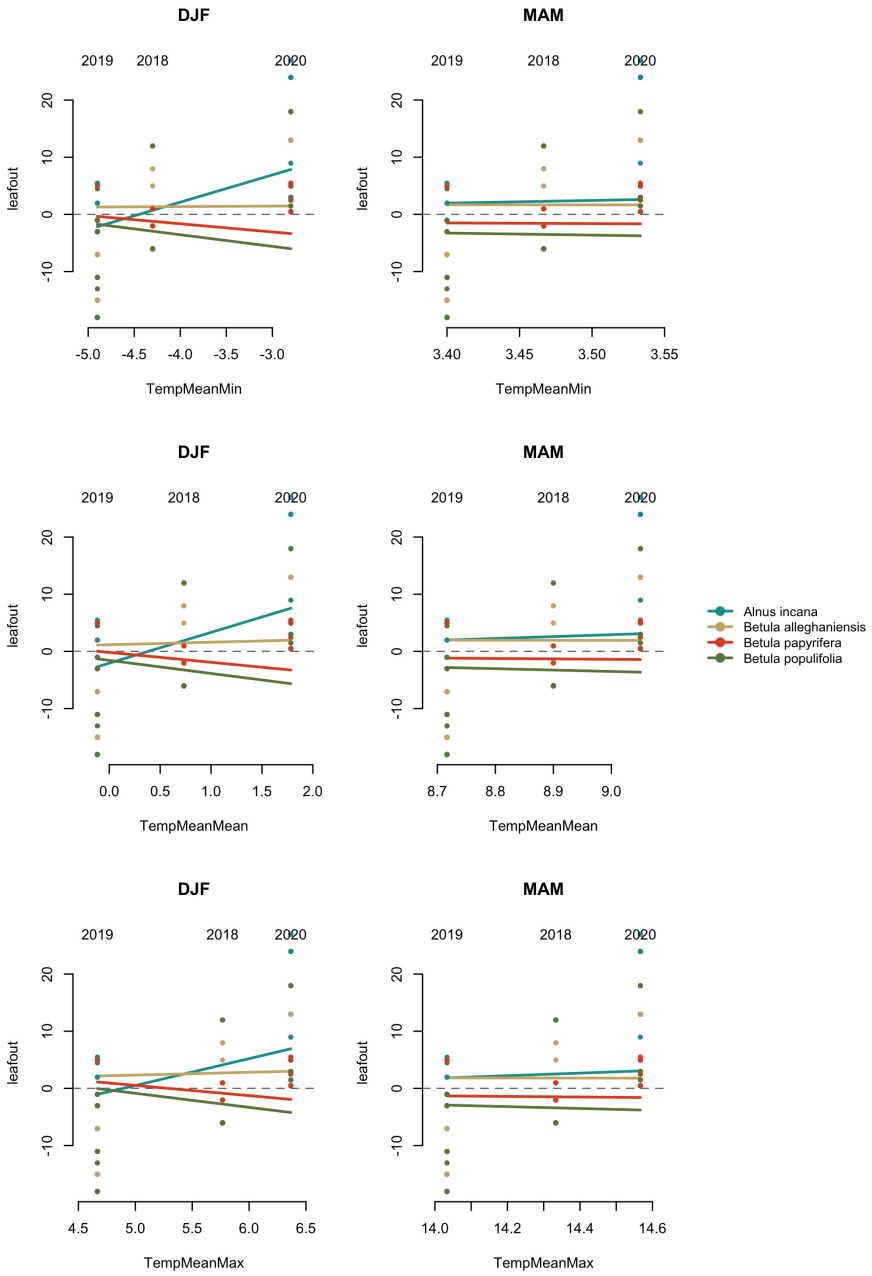

Figure S2: Leafout as a function of different climate predictors across winter (DJF) and spring (MAM). DJF is for December, January and February and MAM is for March, April and May. PDSI represents the Palmer Drought Severity Index averaged for each period. TempMeanMax, TempMeanMean and TempMeanMin are the averaged maximum, mean and mininimum monthly temperature over the 3-month periods. Precip is the total precipitation (mm) received over the 3-month periods.

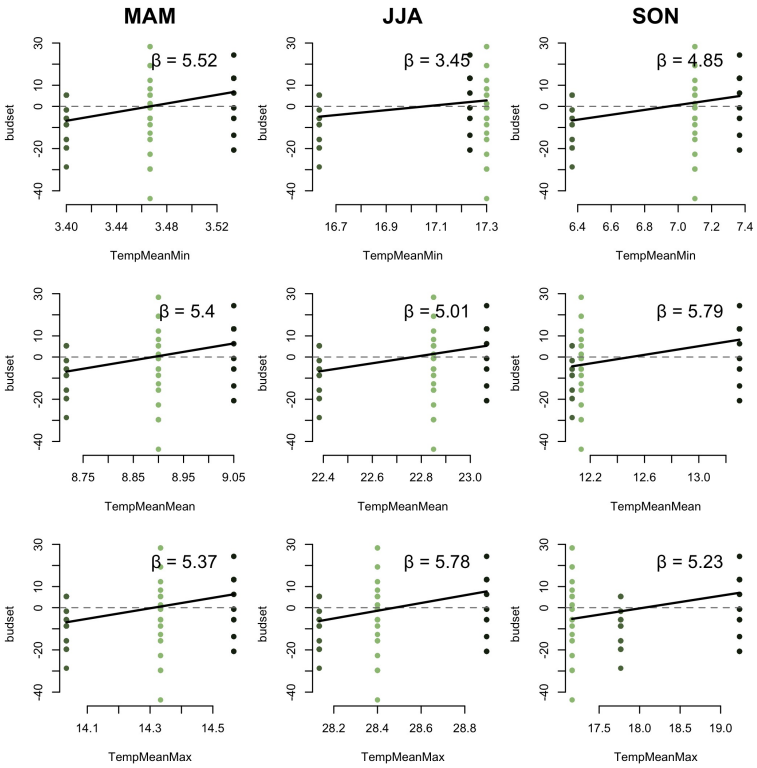

Figure S3: Budset as a function of different climate predictors across spring (MAM), summer (JJA) and autumn (SON). MAM is for March, April and May, JJA is for June, July and August and SON is for September, October and November. PDSI represents the Palmer Drought Severity Index averaged for each period. TempMeanMax, TempMeanMean and TempMeanMin are the averaged maximum, mean and mininimum monthly temperature over the 3-month periods. Precip is the total precipitation (mm) received over the 3-month periods.

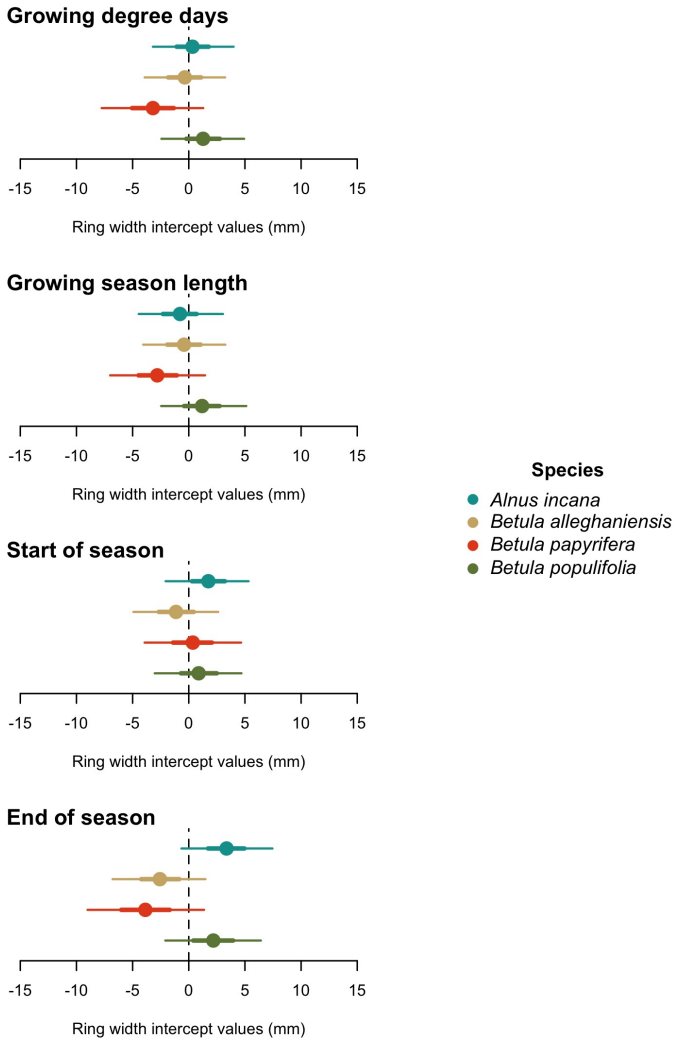
Growing degree days
-15
-10
-5
0
5
10
15
Ring width intercept values (mm)
Growing season length
-15
-10
-5
0
5
10
15
Ring width intercept values (mm)
Start of season
-15
-10
-5
0
5
10
15
Ring width intercept values (mm)
End of season
-15
-10
-5
0
5
10
15
Ring width intercept values (mm)
Species
Alnus incana
Betula alleghaniensis
Betula papyrifera
Betula populifolia

Figure S4:

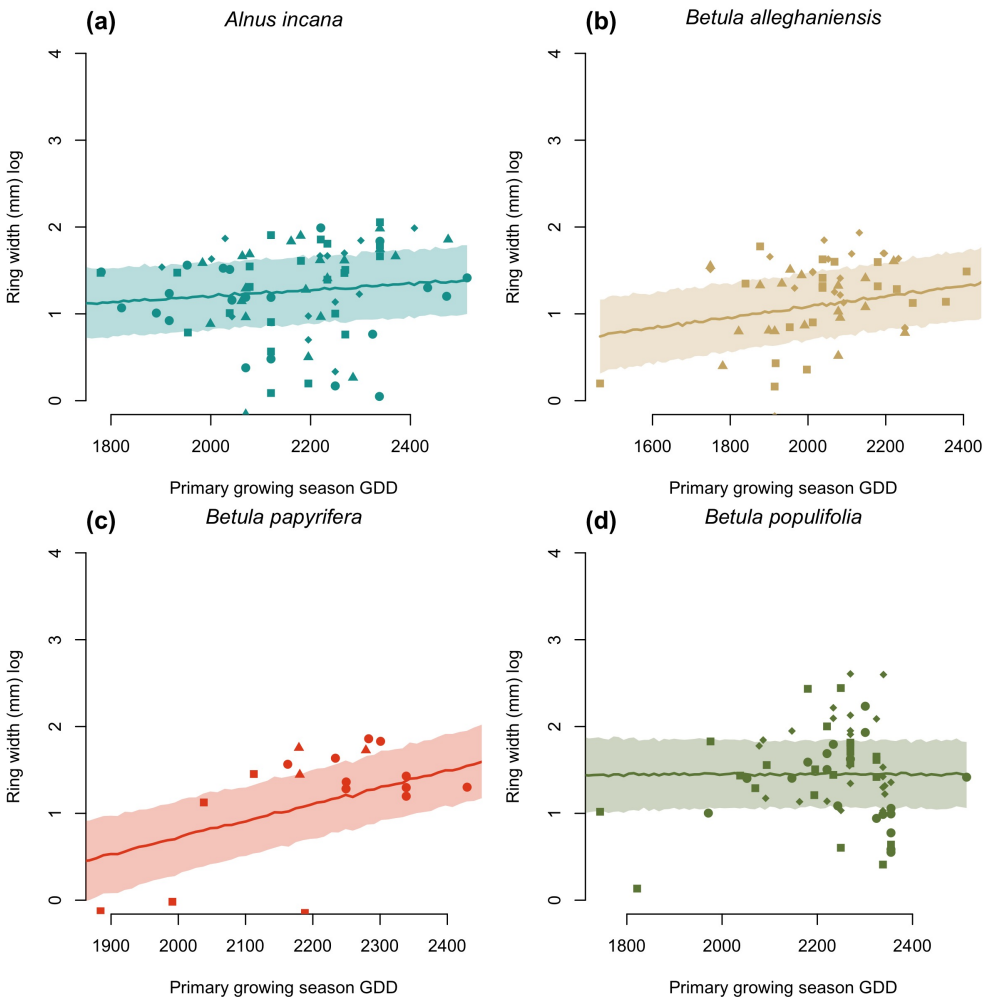

Figure S5: Ring width responses to total growing degree days (GDD) accumulated between the leafout and budset date. The length of the growing season is calculated between the leafout and budset date. The shaded areas represent the 50% confidence intervals of the posterior distribution with measurement error, and the lines are their means. The dots represent the empirical data

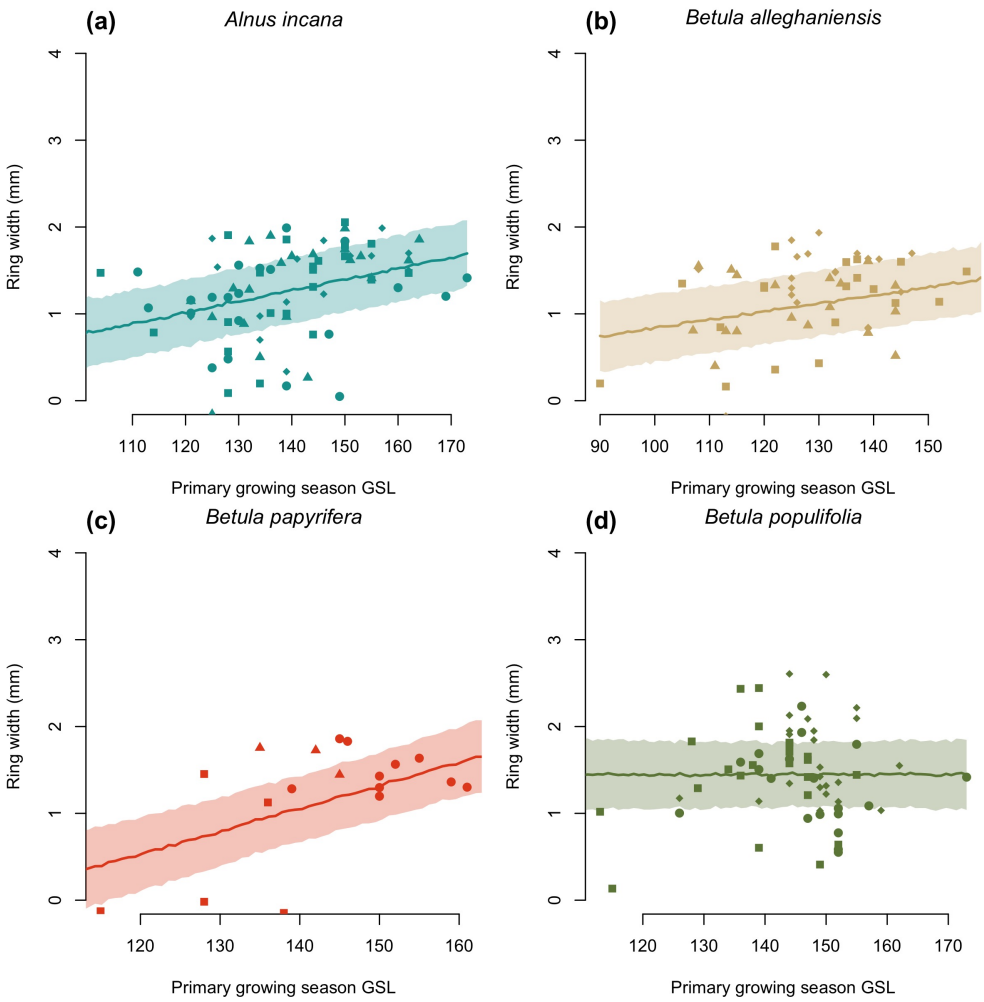

Figure S6: Ring width responses to the length of the primary growing season (days). The length of the growing season is calculated between the leafout and budset date. The shaded areas represent the 50% confidence intervals of the posterior distribution with measurement error, and the lines are their means. The dots represent the empirical data